

Which Way Does the Condition Point?

Unit 2 · Topic 2.6 · THE PROBLEM

Suppose a school screens 1,000 students for a hypothetical condition. Exactly 5% have the condition. The test has 90% sensitivity: among students with the condition, 90% test positive. Its false-positive rate is 10%: among students without the condition, 10% test positive. The implied counts are shown.

Mina: “The test catches 90% of students who have the condition, so a positive result means a student has a 90% chance of having it.”

Luis: “We need to count everyone who tested positive before finding the chance of having the condition.”

Screening results (counts)

Status	Pos.	Neg.	Total
Condition	45	5	50
No condition	95	855	950
Total	140	860	1,000

FIRST TAKE — one sentence before you work the parts

Does a positive result mean a 90% chance of having the condition? Choose a claim and cite one table count.

- DOK 1** (a) For $P(\text{condition} \mid \text{positive})$, identify the numerator and denominator counts.
- DOK 2** (b) Find $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$. Show each count fraction and percent.
- DOK 3** (c) Is Mina’s claim supported? Use the positive-test counts for students with and without the condition. Explain how reversing given changes the group in the denominator. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.